

Constructing micro-dynamic surface of boron-polyurethane with enduring anti-fouling properties

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ABSTRACT

Self-polishing coatings (SPC) with renewable surfaces are widely used in the shipping industry and offshore installations to prevent unexpected costs and safety concerns from marine biofouling. However, self-polishing coatings still suffer from uncontrollable release of antifouling biocides which is called “initial burst and final decay” and poor mechanical properties. Herein, a series of controlled-release boron-polyurethane antifouling coatings (DBIT) were prepared with good mechanical properties and micro-dynamic surface. 2, 2-bis(hydroxymethyl)propionic acid (DMPA) and pyridine-diphenylborane (PDB) were used as reaction sites and antifouling functional side chain groups, respectively, which endowing the coatings with the controlled release ability and good performance. The results showed that the DBIT coatings had certain mechanical properties, up to an ultimate stress of 14.8 MPa. The DBIT₇₅ coating could reduce 90.6 % settlement of *Halimphora* sp., 92.7 % settlement of *Nitzschia closterium*, and repel the settlement of mussels. More importantly, DBIT_x coatings displayed excellent antifouling performance for at least 4 months in the field test, which shows that the controlled-release polyurethane coatings provide a new choice for marine antifouling applications.

1. Introduction

Marine biofouling is viciously accumulated on ships, offshore drilling platforms, causing huge economic losses and damage to the ecosystem [1–3]. Therefore, the prevention and control of marine biofouling are urgent. At present, one of the most economical and effective methods is antifouling paint [4–6]. Acrylate copolymer, polyurethane and polydimethylsiloxane are commonly used as resin matrix of antifouling coatings [7,8]. Among them, polyurethane (PU) materials are widely applied in military and national economy due to their excellent mechanical properties, and stable chemical structure. However, it is undeniable that polyurethane is restricted in antifouling coatings industry in view of its unsatisfactory antifouling properties. To meet the requirements of high-efficiency antifouling, modified PU antifouling coatings have emerged, which is an effective mean to improve the cost performance of PU materials [9].

To enhance antifouling performance, resins are blended [10–13],

immobilized [14–16], microencapsulated with metal-containing antifouling actives [17], or composited by the shell-core structure metal nanorod [18,19]. The above-mentioned metal particles will be undoubtedly released into ocean during application, which is contrary to the principle of sustainability. Also, the addition of photocatalysts [20], non-metal antifouling agents [21], natural antifouling agents [22], and antibacterial compounds [23] is difficult to avoid the “initial burst and final decay” effect [17]. So the strategy of immobilizing the antifouling actives in the resin is better to solve such disadvantage, thereby extending the life of the antifouling coating.

Recently, many antifouling polymers with antifouling groups have been reported. Examples include the acrylate, silicone and polyurethane modified by natural products [24–28], antibacterial compounds [29,30], and commercial antifouling actives [31,32]. The zwitterionic polymers, polyethylene glycol (PEG), etc., are also used to improve the antifouling effect of resins in antifouling coating [33–39]. In previous study, eugenol and coumarin were immobilized in polyurethane as

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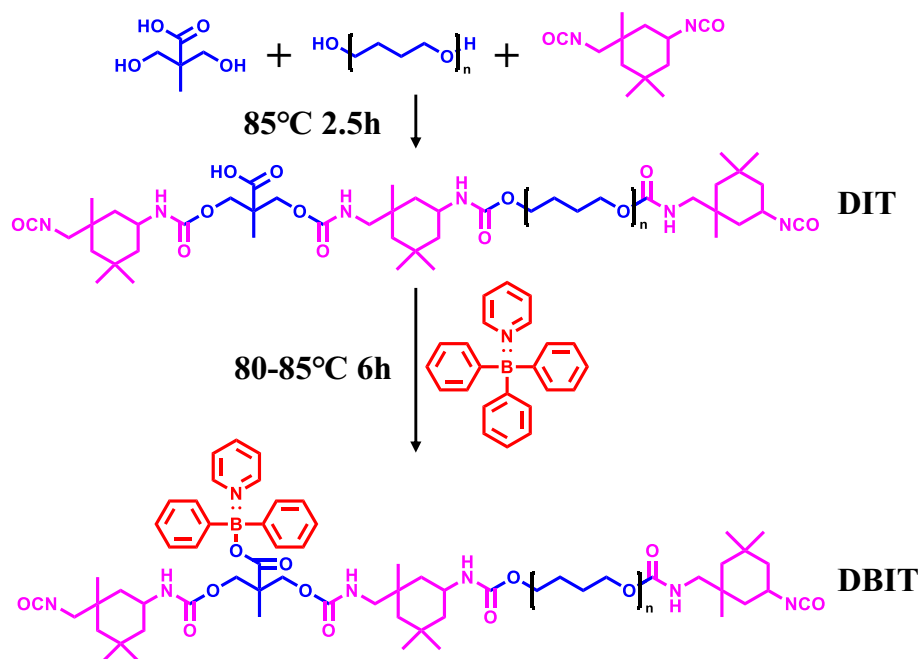


Fig. 1. Synthetic schematic of DBIT pre-polymer by one-pot method.

antifouling actives, and excess coumarin was dimerized to enhance antifouling properties [27]. Ambiguous polyurethane surfaces were formed by both hydrophilic and hydrophobic fluorine segments, and antifouling abilities were exhibited in the marine test [35].

In our previous research [40], the self-polishing acrylate copolymer with pyridine-diphenylborane (PDB) side chains was synthesized and antifouling abilities was verified. In this work, a controlled-release boron-polyurethane antifouling coatings (DBIT) with a dynamic surface were synthesized by immobilizing the organic antifouling functional group PDB in the polyurethane resin. DBIT_x coatings realize a micro-dynamic surface and achieve environmentally friendly. Moreover, the DBIT_x coatings grafting antifouling functional groups showed excellent antifouling properties in diatom settlement, mussel settlement assay and marine field test. Both excellent antifouling performance and mechanical properties of DBIT are compared with the work already reported (Table S1), and the DBIT_x provide an ecofriendly antifouling coating for large-scale practical marine antifouling applications.

2. Experimental section

2.1. Materials and reagents

Isophorone diisocyanate (IPDI, 99 %, Yantai Wan Hua Polyurethane Co., Ltd.), 2,2-bis(hydroxymethyl)propionic acid (DMPA, 98 %, Shanghai Macklin Biochemical Co., Ltd.), butanediol (BDO, 99 %, Shanghai Aladdin Biochemical Technology Co., Ltd.) and pyridine-triphenylborane (PTPB, 98 %, Changzhou Kean Chemical Technology Co., Ltd.) were used as receive. Polytetrahydrofuranediol (PTMG, Mw = 1000 g·mol⁻¹) was obtained from Jining Huakai Resin Co., Ltd. and dried under vacuum in 120 °C to remove moisture before use. Xylene (99 %) and tetrahydrofuran (THF, 99 %) were obtained from Tianjin Fuyu Fine Chemical Co., Ltd. and dried with 4 Å molecular sieves before use.

2.2. Synthesis of DIT pre-polymer and DBIT pre-polymer

The polymer synthesized by the pre-polymer method using IPDI and PTMG as the structural segment and DMPA as the functional segment was named DIT. Typical synthetic schematic is showed in Fig. 1. First,

Table 1

Ingredients of DIT and DBIT_x pre-polymer and crosslinker of coatings.

Sample	DMPA (mol)	PTMG1000 (mol)	IPDI (mol)	PTPB (mol)	BDO of 20 g pre-polymer (mmol)
DIT	0.05	0.1	0.2	–	1.72
DBIT ₂₅	0.05	0.1	0.2	0.0125	1.70
DBIT ₅₀	0.05	0.1	0.2	0.025	1.68
DBIT ₇₅	0.05	0.1	0.2	0.0374	1.66

25 g of THF was added as a solvent in a four-necked flask equipped with nitrogen inlet, mechanical stirring, reflux device and thermometer. Then 6.7 g of DMPA and 100 g of PTMG were added, and 44.4 g of IPDI was added dropwise under stirring at 50 °C about 2.5 h until the solution was clear and transparent. The DIT pre-polymer was obtained.

On the base of DIT pre-polymer, DMPA on the polymer chains was reacted with PTPB, and this pre-polymer was named DBIT. DBIT_x was obtained by the one-pot method as shown in Fig. 1 (x = 25, 50 and 75, represent the mole feeding ratio of PTPB to DMPA). After the DIT pre-polymer was obtained, a certain amount of PTPB and 100 g of xylene were added into the same flask, and then 6 h of reaction was carried out under 80–85 °C. The unreacted PTPB was filtered to obtain light yellow pre-polymers. Table 1 shows the main ingredients of DIT and DBIT_x pre-polymer by feeding mole.

2.3. Preparation of DIT and DBIT_x coatings

Typically, 20 g of pre-polymer and a stoichiometric amount of BDO (Table 1) were used as crosslinker to be added to a rapid stirrer and stirred for 5 min, and the chain extension ratio was about 1.05 (R = 1.05). Subsequently, the mixtures were poured into a mold or coated on glass slide by spin-coating, vacuum cured at 50 °C for 2 h to volatilize the solvent, and then cured at 80 °C for 24 h in drying oven. After curing, a series of coatings were obtained and assays were performed after one week of storage at room temperature.

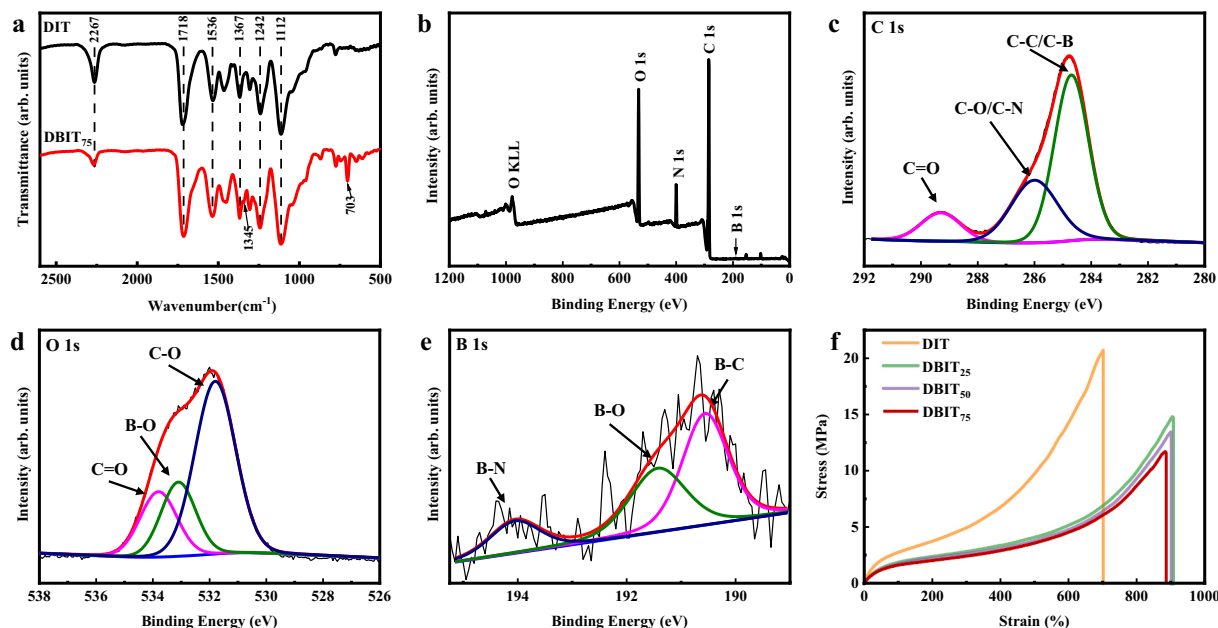


Fig. 2. Chemical structure and mechanical properties of DBIT coatings. (a) FTIR spectra of DIT and DBIT₇₅ pre-polymer. (b) XPS wide-scan spectra of DBIT. (c-e) High-resolution spectra of C 1s, O 1s, B 1s. (f) Stress-strain curve of DIT and DBIT_x coatings.

2.4. Characterization

Details were listed in Supporting Information.

3. Result and discussion

3.1. Chemical structure and mechanical properties of DBIT_x coatings

The reaction of DIT pre-polymer and grafting of PDB groups were confirmed by Fourier transform infrared (FTIR) and X-ray photoelectron spectroscopy (XPS). As shown in Fig. 2a, the characteristic absorption peaks appear in DIT and DBIT₇₅ pre-polymer near 2267 cm⁻¹, 1718

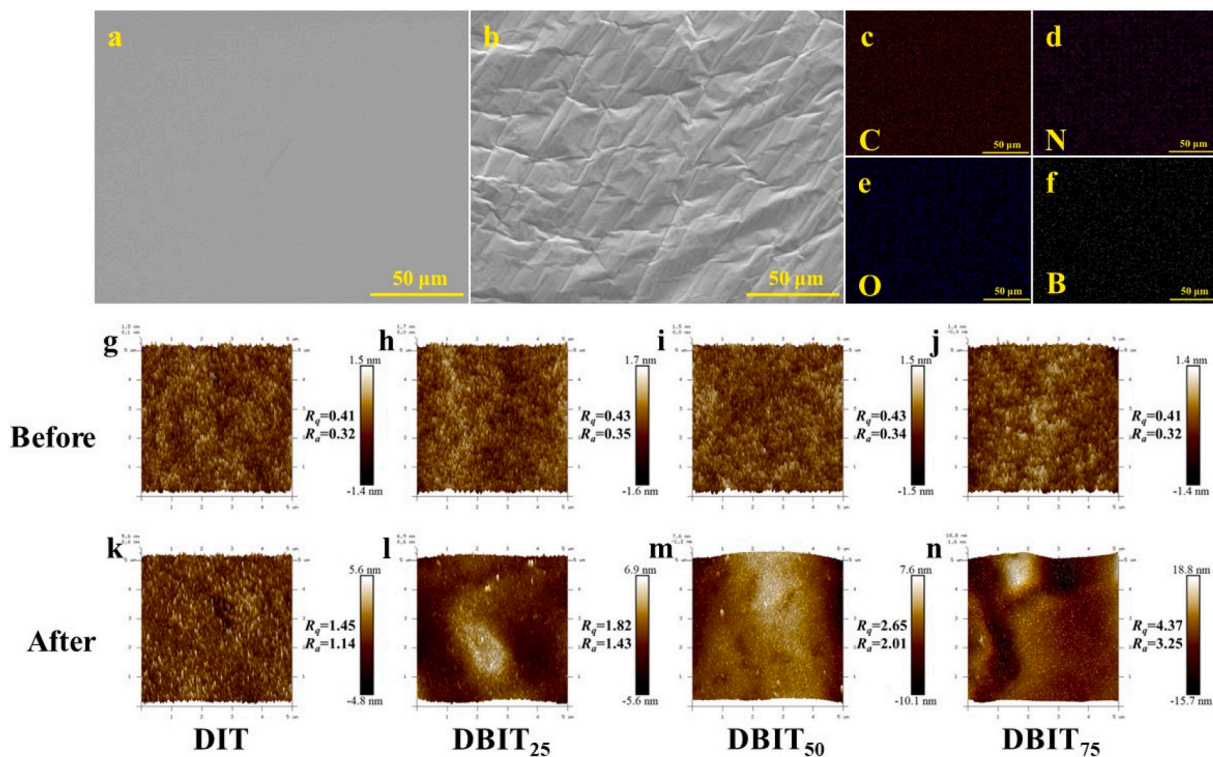


Fig. 3. Topography of different coatings and elements distribution of DBIT₇₅ coating. (a) DBIT₇₅ coating SEM images before immersion, (b) after immersion. (c-f) EDX mapping of DBIT₇₅ coating before immersion. (g-j) AFM images of DIT and DBIT_x coatings before immersion, (k-n) after immersion.

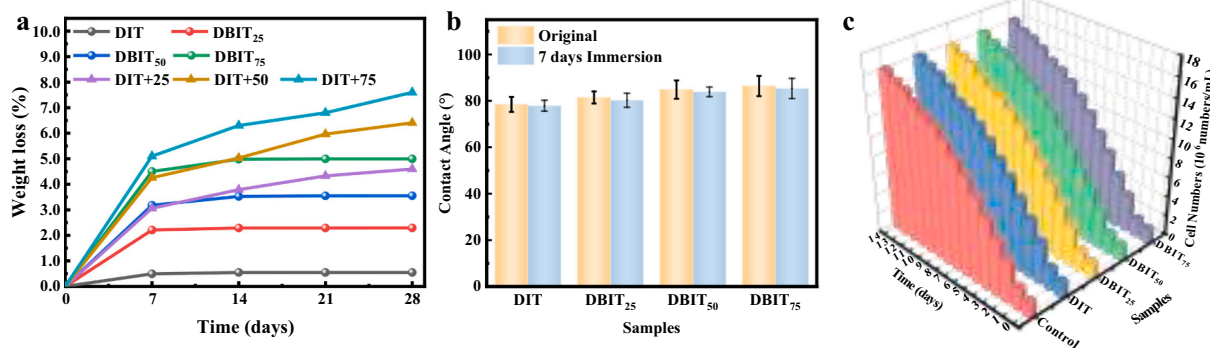


Fig. 4. Underwater stabilities and environment friendly test of DBIT_x coatings. (a) Weight loss test by immersing in ASW (immobilized and non-grafting coatings). (b) Water contact angle of coating surfaces before and after immersion in ASW. (c) Growth curve of *Nitzschia closterium* with different coatings.

cm^{-1} , 1367 cm^{-1} , 1112 cm^{-1} , which correspond to the -NCO asymmetric stretching vibration, the ester carbonyl stretching vibration, the -NCO symmetric stretching vibration and the ether bond stretching vibration, respectively. Moreover, the peak at 1536 cm^{-1} and 1308 cm^{-1} are related to amide II and amide III bands of typical urethane bonds in PU pre-polymers, which indicates the successfully synthesis of pre-polymers. It is worth noting that the B—O bond at 1345 cm^{-1} and the aromatic ring of the PDB group at 703 cm^{-1} found new peaks in DBIT₇₅ pre-polymer, which proves the grafting reaction of boron into the DBIT₇₅ pre-polymer.

To accurately capture the valence state of elements in the DBIT coating, the tested coating has been rationally optimized. Fig. 2b shows the wide-scan spectra of DBIT, which reveals that it contains C, N, O and B elements and element contents are listed in Table S2. In the high-resolution C 1s spectrum (Fig. 2c), three peaks of C-C/C-B, C-O/C-N and C=O are found at 284.8 eV , 286.0 eV and 288.3 eV . Moreover, for O 1s spectrum (Fig. 2d), the three peaks locate at 533.8 eV , 533.1 eV and 531.8 eV are related to C=O, B—O and C—O bond, respectively. And three peaks are identified as B—N, B—O and B—C bond in B 1s spectrum (Fig. 2e) [41,42], respectively. According to FTIR and XPS spectroscopy, PDB group was successfully grafted onto DIT pre-polymers, indicating that DBIT_x pre-polymer were obtained by one-pot method. To clearly detect the change of element content after immersion in artificial sea water (ASW) by XPS, it is summarized and listed in Table S2. This indicates that the boron-containing PDB groups were reduced during immersion due to hydrolysis.

The mechanical properties are significant values as indicators of the reliability of polyurethane antifouling coatings. Fig. 2f shows typical tensile-strain curves of DIT and DBIT_x, which were examined according to ASTM D 2794 standard. And the tensile strain and stress at break and elasticity modulus are listed in Table S3. DIT has higher tensile strength but lower elongation at break than DBIT owing to the reaction between unreacted -COOH groups and -NCO groups coating increased the hard segment content in DIT. In contrast, with the reaction of PDB and -COOH groups, the excess -COOH groups in DBIT will decrease, resulting in a decrease in crosslinking density and the decreased ultimate stress and elastic modulus. Nevertheless, the ultimate stress and elasticity modulus of DBIT₇₅ coating reaches 11.7 MPa and 2.9 MPa , respectively, which is higher than that of typical PDMS-based fouling release coatings and fully meet the practical antifouling application.

3.2. Surface topography and underwater properties of DBIT_x coatings

The adhesion of marine biofouling is closely related to topography of the material surface, the topography of coatings should be studied by scanning electron microscope (SEM). The topography change and element distribution of DBIT₇₅ coating before and after immersion are shown in Fig. 3a. The surface of the coating before immersion was flat,

on the contrary, a certain microstructure appears on the surface of the coating after immersing in ASW for 7 days (Fig. 3b). EDX mapping was used to study the distribution of grafted boron. It can be seen that C, N, O and B are uniformly distributed on the coating surface before and after immersion without partial accumulation (Fig. 3c-f, Fig. S1), which predicts that the coating antifouling groups are uniform by grafting PDB method.

To investigate the microscopic changes of the coating surface in detail, atomic force microscope (AFM) was used to evaluate the roughness before and after immersion. The surfaces of all coatings are relatively flat with no significant difference in roughness before immersion, and R_q and R_a data (given in Fig. 3g-j) are below 1 nm . After all coatings were immersed in ASW for 7 days, the surface topography and roughness data were shown in Fig. 3k-n. All the coatings immersed in ASW were swelled and the different surface undulations were observed in AFM images. The differences of roughness were increased with PDB content, especially the roughness of DBIT₇₅ coating changed from 0.41 nm to 4.37 nm in R_q and from 0.32 nm to 3.25 nm in R_a . Only the DIT coating had a slight increase in roughness from 0.41 nm to 1.45 nm in R_q and from 0.32 to 1.14 in R_a , which means there are other factors determine surface undulations of coating besides swelling after immersion.

XPS was used to further research the coating before and after the immersion. The element contents and the XPS spectra after immersion were shown as Table S2 and Fig. S2. The peak areas of B—O in the high-resolution spectra of B 1s and O 1s decrease, and the boron content also decrease after immersion. Based on the results of XPS, the hydrolysis mechanism of the PDB group is the breakdown of the B—O bond and the newly formed sodium carboxylate leading to the release of PDB [40]. When the coating is swelled, the hydrophilic segments formed by the hydrolysis of PDB migrate from the sublayer to the surface, and the migration leads to the change of the surface topography [43,44]. So the PDB content increase leading to the topography more obviously changed. In addition, there are more crosslink between -COOH groups and -NCO groups in DIT coating, and less new hydrophilic segments are formed, so that the topography of DIT coating is not too much different. And the topography change realizing the micro-dynamic property of DBIT_x coatings.

To further confirm the micro-dynamic change of the coating, the mass loss ratio of the coating within 28 days was measured which is quantified by the mass loss. As shown in Fig. 4a, for all the PDB groups immobilized coatings, the mass dropped significantly in first week and DBIT₇₅ is the most weight loss in all coatings. Moreover, the surface roughness measured by AFM is related to the weight loss of coatings, the hydrolysis of some PDB groups and the separation of short polymer chains from coatings resulted in rapid mass loss and surface topography change. Then, curves became flat in next three weeks, DBIT₇₅ with a mass loss of 4.99% was the most weight loss in the 28-day test and the

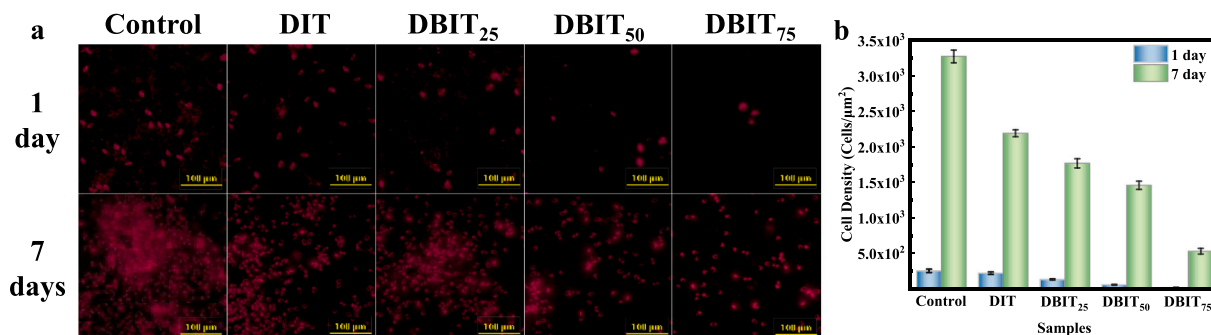


Fig. 5. Diatom settlement assay result of *Halamphora* sp. (a) Fluorescence microscopy images after 1 and 7 days. (b) Diatom cell density.

mass loss ratio is related to the content of PDB groups. Besides, the mass loss ratio of non-grafting coatings and DBIT_x coatings are also shown in Fig. 4a (DIT+25, DIT+50 and DIT+75 represent the coatings with non-grafting method). The coatings synthesized by the non-grafting method produce a phenomenon of uncontrolled rapid release of antifouling actives, the mass loss ratio of the non-grafted coatings was higher than that of the DBIT_x coatings and remained high even after 14 days, which is corresponded to the “initial burst and final decay” effect [17]. Compared with the antifouling coatings without any chemical crosslinking, DBIT series coatings have controlled release ability. The chemically cross-linked PDB antifouling groups can be released after the breakdown of the ester bond by hydrolysis, so the immobilized PDB antifouling groups have lower release rate to meet the controlled release ability.

Surface wettability was also detected by water contact angle (WCA) as shown in Fig. 4b. Initially, the average static water contact angle of DIT was 78.5° and the contact angle of DBIT₇₅ is 86.4°, which increased with the increase of PDB group content. After 7 days of immersion, there were little changes of contact angle among all coatings and the change trends were similar. The reason for the phenomenon can be speculated that there is low molar ratio of PDB groups in the polymer, and the PDB adjacent groups are not strongly hydrophilic groups, the WCA has little effect even after PDB hydrolysis.

Furthermore, to verify the DBIT_x coatings potential environmental issues of the hydrolysis of PDB groups, environmentally friendly test was also carried out and the results are shown in Fig. 4c. After the coatings were immersed in the *Nitzschia closterium* suspend solution for 14 days, the absorbance of the diatom solution were measured by UV–Vis spectrophotometer and transferred to concentration. Diatom concentrations of DBIT_x coatings are not significantly different from the control group, which shows that diatom growth is barely affected by DBIT_x coatings. To some extent, DBIT_x coatings were non-toxic and environmental friendly.

3.3. Antifouling properties of DBIT_x coatings

To explore the impact of surface on antifouling performance of DBIT_x coatings, *Halamphora* sp. and *Nitzschia closterium* were used as the

typical biofouling for the diatom settlement assay. Fig. 5a shows the fluorescence images of the settlement of *Halamphora* sp. after immersion in different coatings for 1 and 7 days. It can be seen that the DBIT_x coatings grafted with PDB groups has less diatoms attachment than the control and DIT coating. The density of attached diatom cells on different samples (Fig. 5b) decreases from 259 cells/μm² to 24 cells/μm² and from 3262 cells/μm² to 534 cells/μm² by 1 and 7 days, as the PDB group increases. Concretely, the diatom settlement inhibition ratio of DBIT₇₅ coating was 90.6 % after 1 day of immersion (Fig. S3a), while the diatom settlement inhibition ratio of DBIT₇₅ dropped slightly to 83.6 % after 7 days. This is due to the reduction of PDB groups and the growth of diatoms. After 14 days of incubation, *Halamphora* sp. adhered to the surface of the coating in a large amount (Fig. S4), but the density of diatoms on DBIT₇₅ was significantly lower than others, and DBIT₇₅ has good resistance to diatom settlement.

In addition, the planktonic algae *Nitzschia closterium*, which has a different habit from *Halamphora* sp., was also selected for the diatom settlement assay. The fluorescence microscopy images of *Nitzschia closterium* and the density of diatom cells are shown in Fig. 6. As the PDB content increased, the diatom cell densities decline from 6459 cells/μm² to 647 cells/μm² and from 11,946 cells/μm² to 847 cells/μm² by 7 and 14 days. For diatom settlement inhibition ratio, that is, 90.0 % at 7 and 92.7 % at 14 days respectively (Fig. S3b). In general, DBIT_x coatings have good performance in resistance to diatoms.

Mytilus edulis was considered as a kind of macro-biofouling, which could do harm to hulls and facilities surface, and it was chosen as model species in this assay. Fig. 7a and Fig. 7b show the position of the mussels in initial and after 24 h on DBIT₇₅ coating and control. Mussels were placed in test area with random initially, and then moved to new place and settled after 24 h, some mussels moved away from test area after the mussels explored and found suitable surface to attach (Fig. 7c-e). Only one mussel attached on the DBIT₇₅ coating and three mussels moved away from DBIT₇₅ for suitable substrates, by contrast, there were four mussels attached on control group. Obviously, DBIT₇₅ coating are not suitable for mussel and has good performance to prevent mussel settlement.

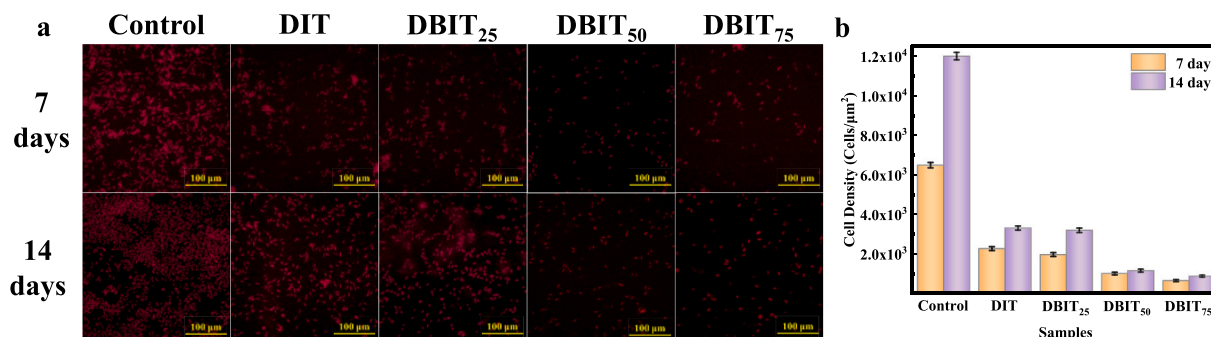


Fig. 6. Diatom settlement assay result of *Nitzschia closterium*. (a) Fluorescence microscopy images after 7 and 14 days. (b) Diatom cell density.

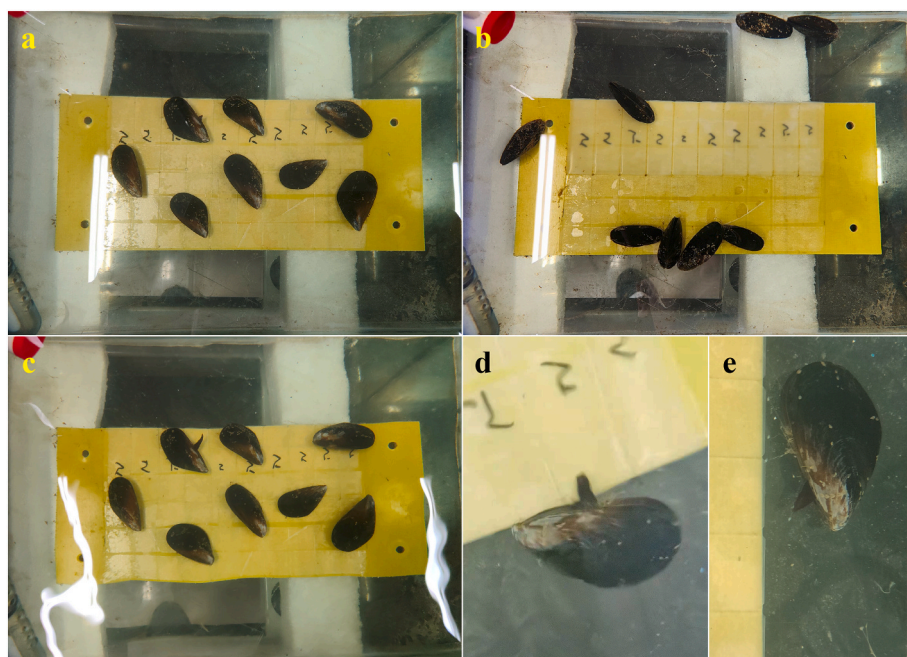


Fig. 7. Digital images during test. (a) Mussels were placed randomly on test area. (b) The location of mussels after 24 h. (c-e) Mussel foot tested surface in different period (DBIT₇₅ in the upper area and control group in the lower area).

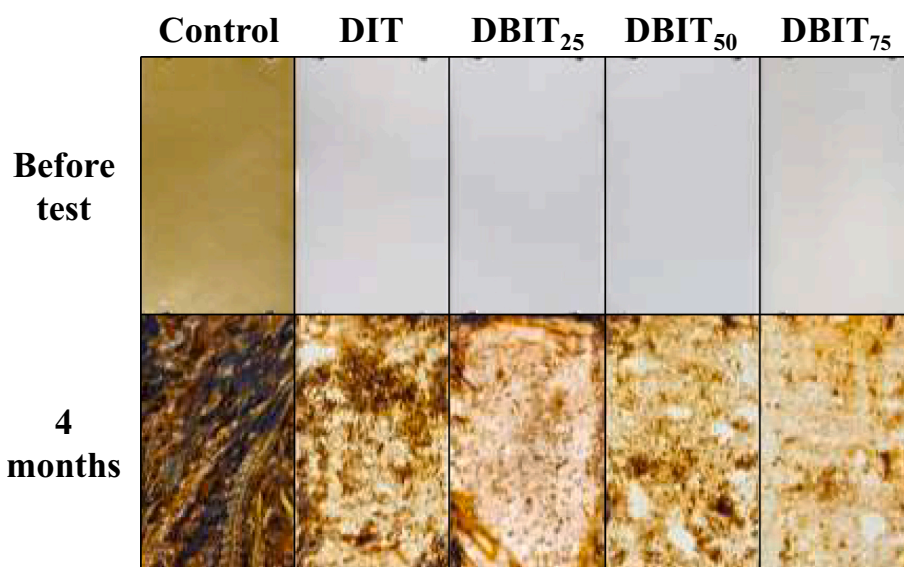


Fig. 8. Digital images of DBIT_x coatings and control plate in marine field test located in the Yellow Sea of China (started from Nov. to Feb. of second year). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Marine field test was carried out to examine the antifouling activities in real sea to simulate the paints in actually use, and place is in the Yellow Sea of China. Uncoating plate made by glass fiber reinforced epoxy was used as control, different coatings (DIT and DBIT_x) were tested and digital images were shown in Fig. 8. In four months, control plate was covered by variety large fouling seaweeds and hard to identify uncoating surface. DIT coatings were covered by small size biofouling, and DBIT_x coatings were less attached by fouling organisms as the PDB groups content increase. That phenomena are consistent with diatom settlement assay. All the DBIT_x coatings displayed good activities, especially, the antifouling activities of DBIT₇₅ coating is better than other samples. The PDB groups actually have positive effect on antifouling according to marine field test.

4. Conclusion

Controlled-release boron-polyurethane antifouling coatings (DBIT) with different addition of PDB antifouling groups were prepared by one-pot method. Compared with the DIT coating, even if the PDB group reduced the mechanical properties, the DBIT₇₅ still achieved 11.7 MPa in ultimate stress and 2.9 MPa in elasticity modulus. According to SEM result, the DBIT₇₅ coating was found appeared undulations after immersion, and further observation was carried out using AFM for all coatings. The coating became rougher because of PDB groups hydrolysis and segment migration, which realizes the micro-dynamic surface. Moreover, antifouling properties had less influence by coarse surface topography, and the DBIT_x coatings performed well in diatom

settlement assays, especially DBIT₇₅ showed diatom settlement inhibition ratio as high as 90.63 % and 92.68 % in *Halimphora* sp. and *Nitzschia closterium* solution, respectively. In addition, DBIT series coatings exhibited environmental friendly by immersing in *Nitzschia closterium* solution. Marine field test displayed excellent antifouling activities of the DBIT_x coatings, especially DBIT₇₅ coating was better than the other. Summarily, DBIT are a series of controlled-release antifouling coatings with potential using areas.

CRediT authorship contribution statement

Jiawei Tang: Conceptualization, Methodology, Investigation, Software, Writing – original draft. Rongrong Chen: Supervision, Writing – review & editing. Xing Huo: Data analysis assistance. Gaohui Sun: Data curation, Resources. Jing Yu: Data curation, Resources. Rumin Li: Data curation, Resources. Peili Liu: Project administration. Qi Liu: Supervision, Writing – review & editing. Jun Wang: Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.porgcoat.2022.107243>.

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